SAMEER SINGH

ssinghzh211@gmail.com 7489215211 linkedin.com/in/sameer-singh-5a74a9379

github.com/Zenith-ss

PROFESSIONAL SUMMARY

Electrical and Electronics Engineering undergraduate with hands-on experience in embedded systems, robotics and Edge AI. Experienced in ESP32/Raspberry Pi prototyping, YOLO model fine-tuning, application development and hardware-software integration through technical projects and hackathons.

ACHIEVEMENTS

Special Recognition – ₹10,000 team prize, *Navomesh 2026* 01/11/2026

2nd Runner-Up – Obstacle Car Race, *Ignition 2.0* 18/09/2026

36-Hour National Hackathon Participant, *Prayatna 3.0* 28/02/2026

Selected in Institute-Level Internal Round – SIH 2026, *Smart India Hackathon 2026* 09/09/2026

Projects

Flint Edge AI Device, *Raspberry Pi 5, YOLO, Edge AI*
• Developed a Raspberry Pi 5-based edge AI prototype integrating computer vision capabilities.
• Fine-tuned a YOLO model initially trained by a teammate and contributed to its integration into the system.
• Worked on hardware integration, enclosure design, system prototyping, and power/thermal management.

EZseva – AI-Powered Healthcare Portal, *Kotlin, Jetpack Compose, Material 3, Gemini API*
- Contributed to UI/UX design, application integration and testing of an Android healthcare platform supporting appointment booking, medical records and hospital selection.
- The platform featured Gemini-powered symptom-to-department recommendations and English/Hindi support.

ESP32-Based Obstacle Racing Car, *ESP32*
• Designed and assembled an ESP32-based robotic car for an obstacle racing competition.
• Implemented wireless control and integrated the hardware for obstacle-course operation.

Real-Life Leveling – Gamified Productivity App, *Flutter, Dart, SharedPreferences*
• Developed a gamified productivity application using Flutter and Dart as an independent learning project.
• Implemented task progression and local data persistence using SharedPreferences.

SIH 2026 – Frontend & UI/UX Prototype, *Frontend Development, UI/UX*
• Contributed to frontend development and UI/UX design for the team's Smart India Hackathon 2026 prototype.
• Worked on interface design and user experience as part of the project team.

Technical Skills

AI & Computer Vision:
YOLO model fine-tuning, model integration, Edge AI prototyping

Embedded Systems
ESP32, Arduino, Raspberry Pi 5

Tools & Development:
Arduino IDE, Git, GitHub, Vercel

Programming & App Development:
C, Dart, Flutter

Hardware Integration:
Sensor interfacing, motor control, relay modules, LCD interfacing

Design & Prototyping:
UI/UX design, rapid prototyping

EDUCATION

B.Tech. — Electrical & Electronics Engineering (EEE), *Bhilai Institute of Technology* Raipur
CSVTU 3rd Semester CGPA: 8.76/10